Jared Lorince, PhD

AI Engineer

jared.j.lorince@gmail.com
jlorince.github.io
linkedin.com/in/jaredlorince
Chicago, IL

SUMMARY

AI engineer with 9+ years spanning academic research, early-stage startups, and large-scale platform engineering. Track record of identifying critical gaps in AI systems, building the foundational infrastructure to close them, and unifying features as they scale from prototype to company-wide adoption. Founding engineer and technical lead, equally at home in framework-level architecture and shipping user-facing product.

CORE COMPETENCIES

AI/ML systems — End-to-end development of LLM-backed product features: provider APIs, prompt engineering, evaluation frameworks, tool-calling and validation infrastructure, latency optimization, and telemetry for measuring AI impact in production.

Software engineering — Full-stack product and platform development across startup and enterprise scale, with a focus on API design and the boundary between framework infrastructure and user-facing features.

Technical leadership — Founding-engineer and tech-lead experience: architecture ownership, cross-team standardization, mentoring junior engineers, and directing contract engineers.

Research & analysis — Experimental design, statistical analysis, and large-scale data processing applied to product and modeling decisions.

Technical skills — TypeScript, Python, SQL, Bash; React, GraphQL and REST API design; PostgreSQL; Git.

PROFESSIONAL EXPERIENCE

Databricks

March 2025 – Present

Senior Software Engineer

- Core contributor to framework and user-facing features for Genie Code, Databricks' AI coding assistant (~80,000 daily active users) — tool selection, diff parsing and rendering, accept/reject flows, and the coordination layer tying disparate AI surfaces into a coherent experience.
- Identified the absence of structured validation in the LLM tool-calling path and designed and built the validation layer (schema enforcement, retry and error-handling logic) now foundational to the internal agent APIs consumed by engineering teams across the ~3,000-person organization.
- Own character-level code-attribution telemetry backing a primary company KPI — the share of code on the platform that is AI- versus human-authored. Significantly expanded an early prototype into the company's source of truth for this metric.
- Standardized how AI-proposed code changes are rendered across product contexts, and drove the design of the integration unifying the inline assistant with the sidebar agent experience.
- Serve as de facto manager for two contract engineers and mentor junior engineers on the team, directing implementation and reviewing work across the AI feature set.

Hex Technologies

May 2021 – March 2025

Staff AI Engineer — Technical lead for a 6-engineer AI team

Sept 2024 – March 2025

- Built a custom SQL parser enabling configuration of no-code data explorations via natural language.
- Built a pipeline that automatically generates suggested prompts from customer data, giving users contextual examples for Hex's AI features.
- Integrated AI typeahead into the product, providing live in-editor code completions.

Senior AI Engineer — Founding engineer of Hex Magic Jan 2023 – Sept 2024

- Sole developer on the initial integration of AI features into the platform.
- Architected an agent-agnostic LLM prompting pipeline underlying all AI features, including custom event-tracking instrumentation.
- Designed and implemented an evaluation framework to quantify prompting quality and enable experimentation with new strategies.
- Built "Magic Analysis," letting users ask natural-language questions and receive an intelligently selected combination of SQL, Python, and visualizations in response.

Senior Software Engineer (10th employee) May 2021 – Jan 2023

- Designed Hex's custom notebook file format, enabling GitHub sync, import/export, and other enterprise features.
- Implemented dataframe SQL; designed and shipped the public API; added R language support.

Narrative Science, Inc. (acq. Salesforce/Tableau) May 2017 – May 2021

Senior Software Engineer → Data Intelligence Technical Lead 2019 – 2021

- Developer on the core rule- and template-based NLG platform (pre-LLM era) generating natural-language narratives from complex datasets.
- Member of the incubation team prototyping high-risk/high-reward features, including automatic extraction of linguistic expressions for abstract concepts from unstructured text (see Patents).
- Served as NLP and ML subject-matter expert.

Software Engineer 2017 – 2019

- Built a custom event-tracking framework enabling stakeholders to analyze user interaction with the platform.

EARLIER EXPERIENCE

Postdoctoral Fellow, Northwestern Institute on Complex Systems (NICO) (2016–2017) — Research in complex systems and computational social science; projects included predicting the emergence of new scientific fields and musical genres, and large-scale data visualization.

Data Scientist, StumbleUpon (2015–2016) — Lead developer of an Apache Spark framework combining topic modeling and rating prediction for user-interest profiling and content recommendation.

Internships: StumbleUpon, Data Science Intern (Summer 2015); Yahoo! Labs, Research Scientist Intern (2011–2012).

EDUCATION

Indiana University, Bloomington 2010 – 2016

Joint Ph.D. in Cognitive Science and Cognitive Psychology

NSF IGERT Fellow. Served as cognitive-science advisor on IARPA's SIRIUS program, designing and analyzing serious-game environments to train intelligence analysts in recognizing and mitigating cognitive biases.

University of California, Berkeley 2005 – 2009

B.A. with High Honors in Cognitive Science

PATENTS

- Bischof, B., **Lorince, J.**, Colgrove, C., McCardel, B., Takahashi, G., & Storr, A. (2025). Priming generative AI model leveraging directed acyclic graph-driven notebook environment. U.S. Patent No. US 12,332,931.
- **Lorince, J.**, Storr, A., McCardel, B., Miller, I., Colgrove, C., & Bischof, B. (2024). Machine learning-assisted code generation in directed acyclic graph-driven notebook environment. U.S. Patent Application No. US 2024/0256228 A1 (pending).
- Smathers, M., Platt, D., Nichols, N., & **Lorince, J.** (2022). Applied artificial intelligence technology for adaptive natural language understanding with term discovery. U.S. Patent No. US 11,341,330 B1.
- Platt, D., Nichols, N., Smathers, M., & **Lorince, J.** (2022). Applied artificial intelligence technology for using natural language processing to train a natural language generation system with respect to numeric style features. U.S. Patent No. US 11,232,270 B1.
- Platt, D., Nichols, N., Smathers, M., & **Lorince, J.** (2020). Applied artificial intelligence technology for using natural language processing and concept expression templates to train a natural language generation system. U.S. Patent No. US 10,706,236 B1.

Plus 9 peer-reviewed papers and book chapters — full list at jlorince.github.io.